# TradePilot — Strategy Research: Profitability, Usability, Platform

*Research-only. Three pillars — make the engines more profitable, make it user-friendly, make it an agentic platform — and the single thread that ties them together.*

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| **Project** | TradePilot (autonomous agentic trading) |
| **Document** | Strategy research — 3-pillar synthesis (no implementation) |
| **Version** | `v1.0.0` |
| **Status** | Research complete |
| **Created** | 2026-06-30 |
| **Method** | 3 parallel research agents (UX · profitability · platform), each grounded in our prior validated findings + external literature |


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## 0. The one finding that ties all three pillars together

The three research threads were run independently. They converged on **the same two truths**:

1. **Simplicity beats complexity — everywhere.** Profitability: a 5-tree model out-hits the 1,735-tree model; execute-at-open beats the rescore loop. UX: a single status surface beats the 10-tab dashboard. Platform: white-box beats black-box. Every pillar's top lever is *remove complexity*, not add it.
2. **The explainable decision pipeline is one artifact with four payoffs.** The "why did it trade X" decision log is simultaneously: the **profitability** validation (auditable, white-box), the **UX** trust surface, the **compliance** answer (white-box avoids the SEBI RA licence + satisfies the audit trail), and the **platform** architecture (the audit/memory agents). Build it once, win four times.

And a structural bonus: the frontier agent architecture (Pillar 3) **prevents the overfitting that's hurting profitability** (Pillar 1) — alpha agents never see evaluation-window returns; memory separates train/eval namespaces. The platform upgrade and the profitability fix are the *same* investment.

## 1. Pillar A — Make the engines more profitable

Every internal finding was independently confirmed by external quant literature. The edge is real (v5 alpha t=3.0, cost-robust to ~45bps) but destroyed at the execution and model-complexity layer. Five levers, ranked:

| # | LEVER | LEVERAGE / EFFORT |
|---|---|---|
| 1 | **Execute at open, kill the rescore loop** — lock prior-close scores, enter 09:15–09:20, hold | Highest · zero cost (code) |
| 2 | **Collapse the ML model to ≤20 trees + purged walk-forward** (embargo folds, 6–8 OOS windows) | High · 2–3 wks |
| 3 | **ATR-proportional sizing, ≥60% deployment** (fractional Kelly on the validated edge) | High · changes return profile now |
| 4 | **Binary regime gate on *direction*** (NIFTY 20-day SMA + VIX-rank → disable shorts on weak days) | Medium · drawdown control |
| 5 | **Retrain ML on the correct label** — P(hit +1.5% before −0.75%), not open→close return | Medium · fixes label/exit mismatch |

**External backing (selected):**

- **Overfitting is existential.** Bailey & Lopez de Prado: after ~4.75 independent backtests, a Sharpe of 1.0 is achievable *by pure chance*. A 1,735-tree model tuned over hundreds of experiments has a meaningless undeflated Sharpe. ScienceDirect (2024): complex models hit ~50% out-of-sample accuracy with "chaotic" hyperparameter sensitivity — the fingerprint of overfit. *Simpler models generalize because they can't memorize enough to fit noise.*
- **Alpha decays with execution lag.** Man Group: a signal's value falls as the lag between generation and execution grows. A morning-momentum signal has a half-life in *minutes*; entering 30–60 min late (the rescore loop) chases exhausted moves.
- **Underbetting destroys returns as surely as overbetting** (Kelly). 35% deployment on a t=3.0 edge leaves half the return on the table; half-Kelly keeps ~75% of growth at half the variance.
- **Regime filters give modest gains, not magic** — and only if kept simple. A filter that fires 15 times in 2 years can't be optimized without curve-fitting. Gate *direction of the book*, not individual entries; two unoptimised signals (SMA + VIX-rank).

**What to AVOID (literature-backed):** more model features (noise amplifiers), optimising regime params, re-expanding to NIFTY-200 without a separate mid-cap edge, leverage to mask under-deployment, a second intraday entry window (churn disguised as diversification), in-sample validation, standard K-fold (leaks future data — use purged folds).

## 2. Pillar B — Make it user-friendly

The reframe: **in an agentic product the user is a supervisor, not a pilot.** The UX job is to make supervision safe, legible, and low-friction — the opposite of a feature-dense builder.

| MOVE | WHAT IT MEANS | WHY |
|---|---|---|
| Lead | **Single status surface** – "is the agent running? what did it do today? anything unusual?" | Replaces 10 tabs; Linear's "is everything okay?" pattern |
| Lead | **Plain-language decision log** per trade ("bought RELIANCE – momentum + volume, 0.5% size, stop 2455") | The product, not cosmetics; also the SEBI audit trail |
| Lead | **Progressive-delegation onboarding** – Observe → Approve-first → Supervised-auto → Full-auto | Users who *earn* full delegation keep it; day-one autonomy gets abandoned |
| Add | **"Why did it do that?"** per-trade explainer; **drawdown/risk** display (not just P&L) | Users tolerate losses they expected, not ones they didn't know were possible |
| Add | **Weekly performance narrative** via Telegram; **progress-vs-Nifty** framing | Less panic-selling than raw daily P&L |
| Cut | The 10-tab nav, the A/B engine toggle, real-time ticker feeds – into a **builder-only mode** | A user watching tickers isn't trusting the agent |

Telegram = **check-not-operate**: end-of-session summary + anomaly alert + intervention-request only. "Timing beats intelligence" – never trade-by-trade noise. Keep the current builder view via a **dual-mode UI** (builder vs user), not a rewrite.

## 3. Pillar C – Make it an agentic platform

**Recommended product shape for India: "BYOB White-Box Agent"** – *your own autonomous AI agent running in your Zerodha/Fyers account; set your mandate once; it researches, decides, and trades; you see every reasoning step.* The platform holds **zero capital** (no PMS/fiduciary licence); **white-box** sidesteps the heavy black-box RA gate. eToro's Agent Portfolios proved the shape; `india-trade-cli` (7-agent OSS, Zerodha+Fyers live) proves it's buildable indie-scale.

**T3 → T4 architecture upgrade** – decompose the monolith into a 9-role agent DAG:

| LAYER | AGENTS | KEY PROPERTY |
|---|---|---|
| Strategy | Planner + **Meta-agent** | Rewrites agent prompts/code from outcomes (the T4 unlock) |
| Signal | Alpha agents | **Never see evaluation-window returns** (anti-overfit) |
| Control | Risk-gate agents | **Binary block**, never modify signals |
| Build | Portfolio, Backtest, Execution | Deterministic solvers; cost/slippage accounting |
| Trust | Audit + Memory agents | **UUID-indexed, immutable, train/eval namespaces** |

**Multi-tenant requirements:** per-user state/ledger/IPS/memory namespace; per-user broker creds (scoped API keys); per-user mandate (Investment Policy Statement) the orchestrator reads before dispatch; audit trail as a product primitive (SEBI: Client → Algo-ID → Static IP → API key, 5-yr retention).

**Phased path:**

| PHASE | WHEN | WHAT | COMPLIANCE |
|---|---|---|---|
| 0 – Personal T4 | now → 6mo | 9-role DAG, meta-agent, UUID memory, explainability log | Zero (paper/own capital) |
| 1 – Invite-only BYOB | 6–12mo | 3–10 trusted users, per-user IPS + broker vault + static IP, white-box algo-ID | Light (white-box, no RA licence) |
| 2 – Closed beta | 12–18mo | 50–200 users, subscription, copy-agent of our own variants | SEBI RA licence, ISO 27001 prep |
| 3 – Platform | 18–24mo | third-party agent SDK + marketplace | Empanelment, VAPT, ISO 27001 |

## 4. The synthesis – what to actually pursue

★ Insight ———————————————— The three pillars are not three projects – they share a spine. **Profitability says "simplify the model and execute clean"; the platform architecture says "decompose into agents whose alpha layer can't see the future"; the UX says "show the user every decision."** All three are satisfied by one build: a **simple, explainable, clean-execution decision pipeline.** Shrink the model (profit), wrap each decision in a logged rationale (UX trust + compliance), and structure it as auditable agents with train/eval separation (platform + anti-overfit). You don't choose between the three pillars – you build the spine and all three light up. ————————————————

**The sequence the research implies (still research – not a build order to execute yet):** 1. **Profit first, cheaply:** Levers 1–3 (execute-at-open, shrink the model, deploy capital) are mostly code + retrain, zero compliance, and prove the engine before anything is productised. 2. **Then the explainable pipeline:** build the decision log / white-box rationale – it's the UX trust surface *and* the compliance artifact *and* the platform's audit layer, all at once. 3. **Then phased productisation:** Phase-0 personal T4 architecture → invite-only BYOB → product, with compliance scaling per user exactly as our SEBI research predicted.

## 5. Open research questions (next, if you want to go deeper)

| # | QUESTION | PILLAR |
|---|---|---|
| Q1 | Capacity curve for v5 – how does slippage scale with lot size (order/30-day-ADV)? Sets the ceiling on capital deployment | Profit |
| Q2 | Does `backtest-honest-fills.py` use purged folds or leaky K-fold? Quantify OOS-Sharpe inflation before trusting the 5-tree "win" | Profit |
| Q3 | Short-book regime regression – on losing-short days, what was NIFTY return + VIX vs median? Confirms the binary regime gate | Profit |
| Q4 | Indian retail-trader interviews (5–10) – what does a non-builder fear about an agent trading their capital? | UX |
| Q5 | Telegram approval-flow latency (P50/P95) during market hours – does approve-first need a fallback default? | UX |
| Q6 | Reconcile the two existing long-only experiments ( `v5_long` vs `tradepilot-v5-longonly-ab` ) | Platform/Profit |